Dead Frequencies

CTF Write-up · Audio Analysis & XOR

This challenge provides an encrypted file and an audio clue. The solution comes from inspecting the audio visually, recovering the key hidden in its spectrogram, and using that key to decode the encrypted text.

Flag

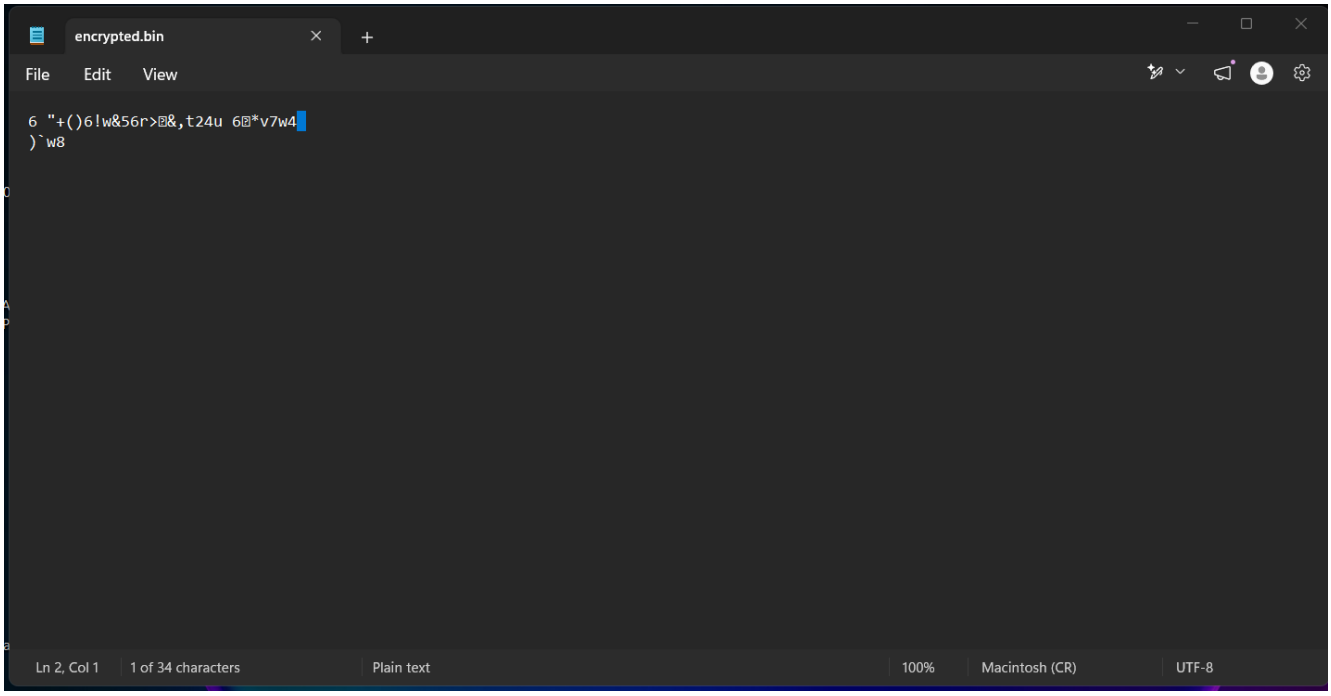
```
recon{sp3ctr4l_wh1sp3rs_n3v3r_l13}
```

The flag shown in the supplied evidence is reproduced above. The steps below document the workflow visible in the screenshots.

1. Inspect the provided files

The challenge includes an encrypted file named **encrypted.bin** and an audio file named **signal.wav**. Opening the encrypted file as ordinary text produces unreadable characters, which suggests that it needs a decoding step rather than simple text cleanup.

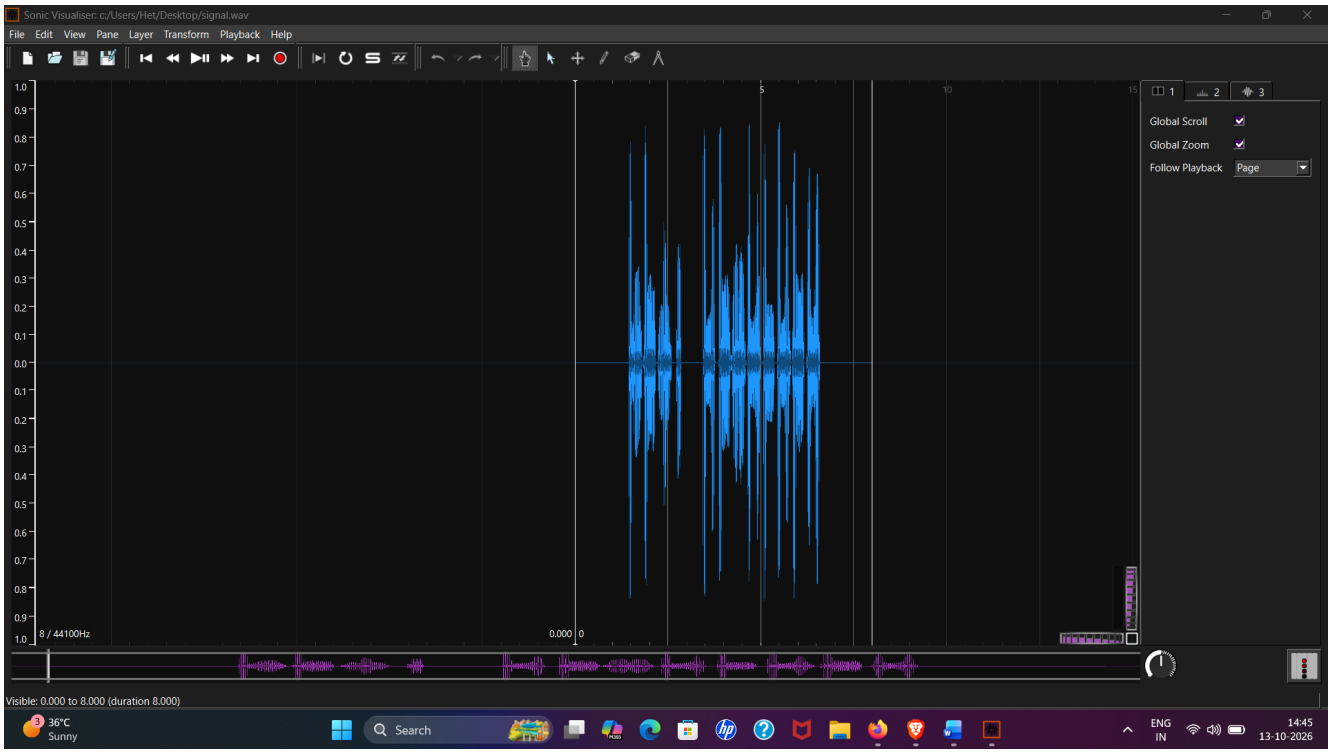
Evidence: encrypted.bin contains non-readable text



2. Examine the audio in Sonic Visualiser

I opened **signal.wav** in Sonic Visualiser and inspected its visual representation. The waveform shows that the file contains several distinct bursts, so I switched to a spectrogram view to check whether information had been encoded in the frequencies.

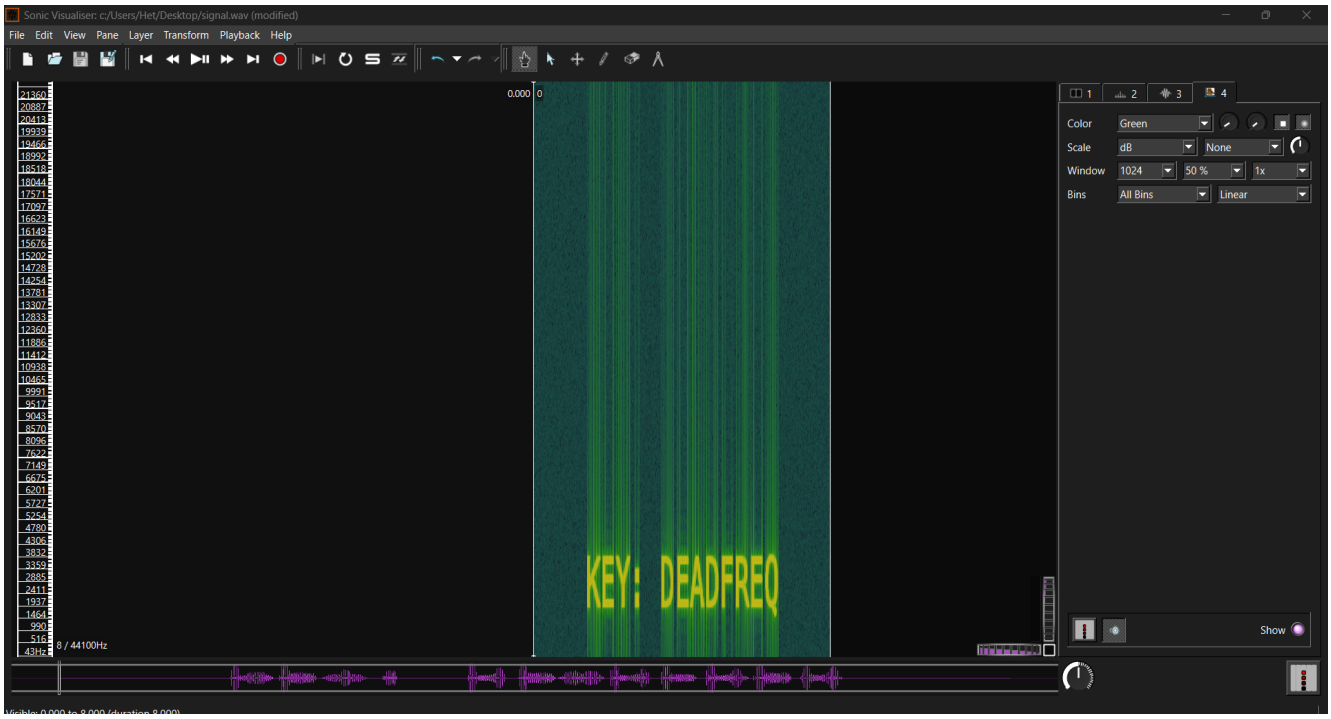
Evidence: audio waveform


Sonic Visualiser: c:/Users/Het/Desktop/signal.wav
File Edit View Pane Layer Transform Playback Help
Global Scroll
Global Zoom
Follow Playback
Page
8 / 44100Hz
0.000
Visible: 0.000 to 8.000 (duration 8.000)
36°C
Sunny
Search
ENG
IN
14:45
13-10-2026

3. Recover the XOR key

The spectrogram reveals the text **KEY: DEADFREQ**. This is the important clue: it provides the key needed for the next step. The audio is therefore not just background material; its visual spectrum carries the decryption hint.

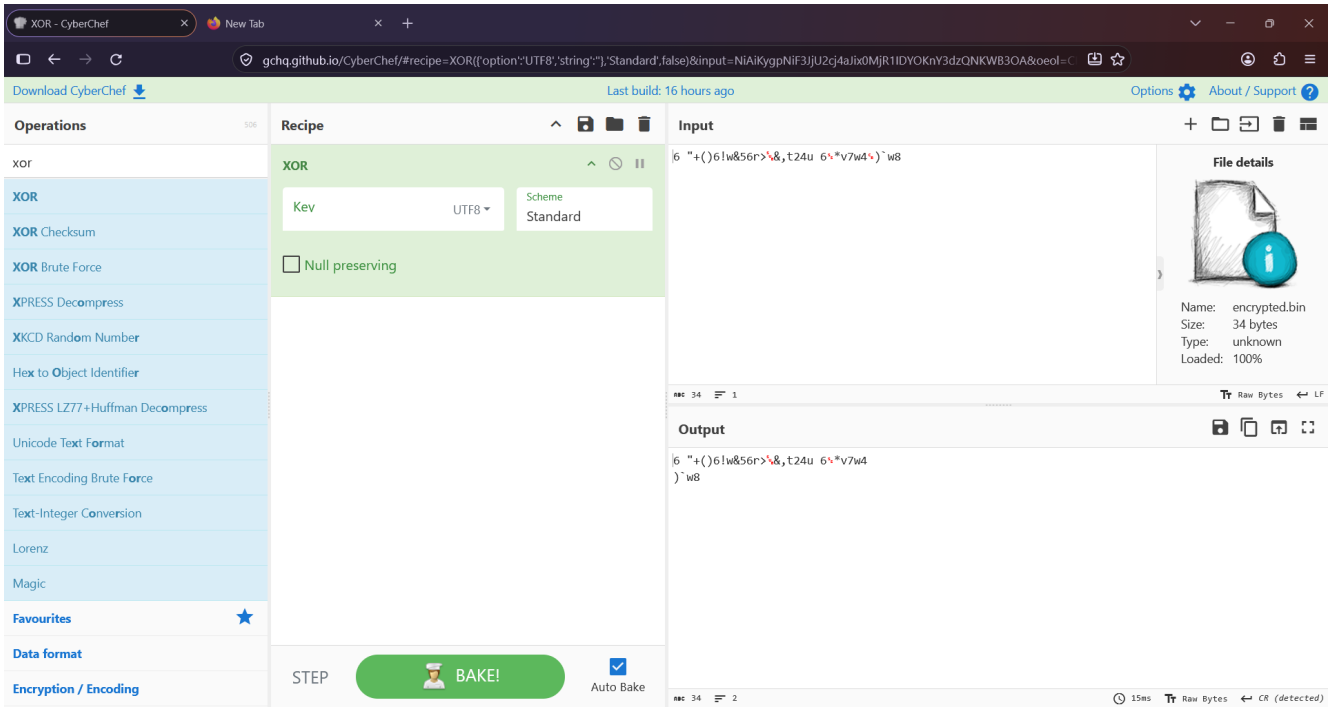
Evidence: spectrogram displaying KEY: DEADFREQ



4. XOR-decode the encrypted content

I used CyberChef with the **XOR** operation. The encrypted content was placed in the input field, and the key was set to **DEADFREQ** with UTF-8 selected. Once the operation was applied, the output became readable and returned the challenge flag.

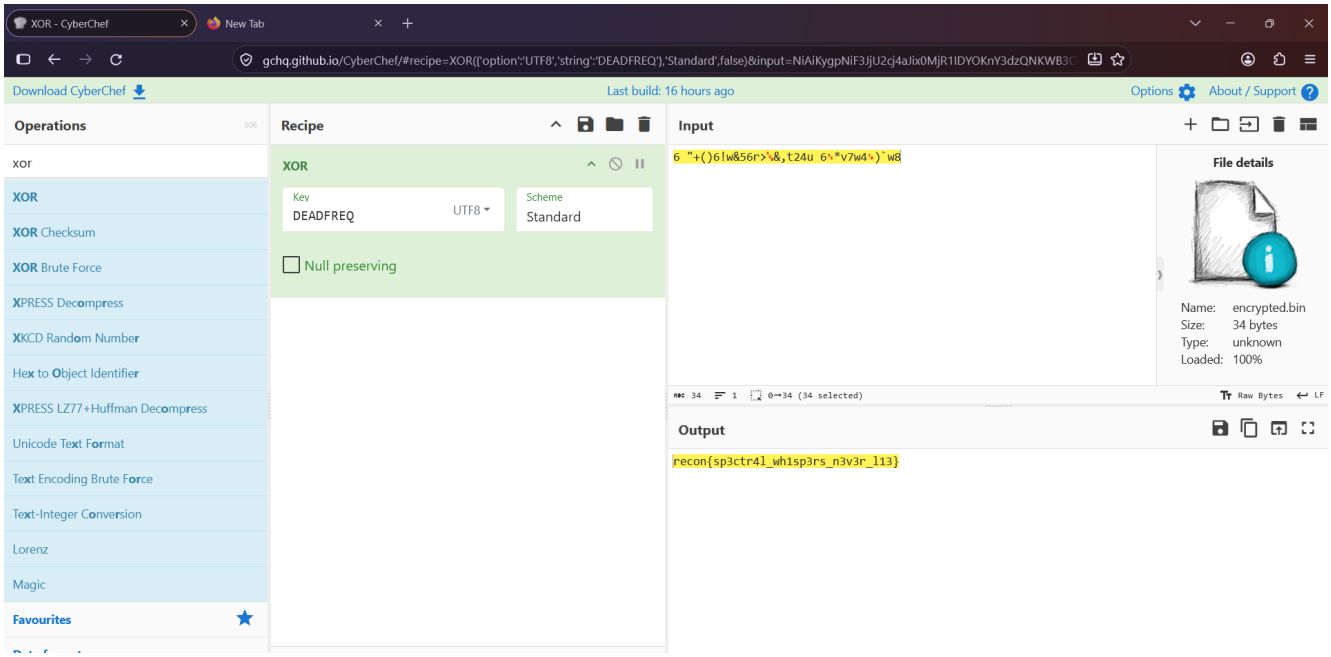
Evidence: XOR operation before applying the key



5. Confirm the result

With the key entered as **DEADFREQ**, CyberChef outputs the flag below. This matches the result shown in the supplied screenshot.

Evidence: CyberChef output



Final flag

```
recon{sp3ctr4l_wh1sp3rs_n3v3r_l13}
```

Tools used: Sonic Visualiser (spectrogram inspection) and CyberChef (XOR).